KL DEEMED TO BE UNIVERSITY • KLH HYDERABAD

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING • DATABASE SYSTEMS & DATABASE ENGINEERING (DBS & DBE)

Corporate Asset Management (CAM): Enterprise CapEx Governance Platform

Problem Statement #28 • Track 4: Enterprise Resource Planning & Financial Intelligence

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1. ABSTRACT

Traditional enterprise resource planning (ERP) database management systems document capital equipment as isolated, static records, causing fatal mismatches between balance-sheet book values, macroeconomic capital inflation, custody trails, and operational utility overheads. This project engineers a resilient, transactional database architecture and unified governance platform entitled **Corporate Asset Management (CAM)**. Utilizing a **MySQL 8.0 (InnoDB)** transactional persistence engine alongside **Spring Boot 3.3.4 (Java 21)**, **React 18**, and **Scikit-Learn**, CAM resolves the legacy unreactive cost crisis. The system models straight-line depreciation against compounding replacement inflation (CapEx Scissors), integrates Bureau of Labor Statistics (BLS) CPI historical indexing (1940–2026), enforces three-tier Role-Based Access Control (RBAC), tracks dynamic loan equity amortization, monitors workforce cost-per-seat ceilings, and captures datacenter Green IT electrical telemetry.

2. PROBLEM STATEMENT & INDUSTRY GAPS

Modern enterprises manage millions in capital hardware yet govern these investments through fractured, disconnected data stores. Problem Statement #28 identifies four systemic flaws in legacy infrastructure:

- **Static Ledgers & Silos:** Hardware records exist as detached rows without referential links to employee custody or departmental headcount budgets.
- **The Inflation Deficit Trap:** Depreciation reserves are booked strictly on historical acquisition cost, leaving organizations severely underfunded when replacement costs climb.
- **Lack of Sovereign RBAC Separation:** Executive real estate and private aviation share identical access boundaries with operational IT hardware.
- **Unreactive Cash Burn:** Maintenance logs, loan EMIs, and datacenter energy footprints are evaluated retrospectively instead of being projected deterministically.

3. SYSTEM ARCHITECTURE & DATA PIPELINE

Layer / Tier	Technology Core	Database & Operational Function
Presentation Tier	React 18 SPA, Vite 5, CSS Grid	Safe CSV preview ingestion, dynamic charts, dual themes.
Application Tier	Spring Boot 3.3.4, Java 21, JPA	REST API gateway, transactional boundary management, CORS filter.
Persistence Tier	MySQL 8.0 Community (InnoDB)	7 normalized relational tables, strict foreign keys, ACID compliance.
Analytical Layer	Scikit-Learn Autoregression	6-month predictive rolling OpEx forecast with $R^2 \geq 0.98$.
Macro Data Layer	Curated BLS CPI Series	Historical purchasing power normalization spanning 1940 to 2026.

4. RELATIONAL DATABASE SCHEMA DESIGN (MYSQL 8.0)

The database `corporate_asset_management` is normalized to 3NF across seven primary tables to maintain strict referential integrity and eliminate update anomalies:

- `departments`: Primary storage for department budgets, ceilings, and headcounts (PK: `id`).
- `employees`: Workforce records linked via foreign key `department_id` to track seat assignments.
- `assets`: Core inventory capturing asset tag, category, book value, procurement cost, power rating, and FKs to `department_id` and `custodian_id`.
- `asset_finances`: Loan financing structures, interest rates, tenures, and monthly EMI commitments (FK: `asset_id`).
- `emi_payments`: Transaction ledger recording historical amortization and principal payoffs (FK: `finance_id`).
- `custody_histories`: Immutable audit log tracking custody handovers between employees (FKs: `asset_id`, `employee_id`).
- `maintenance_logs`: Granular log of repairs and service fees to enforce scrap-versus-repair threshold policies (FK: `asset_id`).

5. MATHEMATICAL & DATABASE FORMULATION MODELS

A. Straight-Line Depreciation

$$\text{Annual Depreciation (\$)} = (\text{Purchase Cost} - \text{Salvage Value}) / \text{Estimated Useful Lifespan (Years)}$$

B. The CapEx Scissors Replacement Deficit

$$\text{Compounded Replacement Cost: } R(t) = C_0 \times (1 + r)^t$$

$$\text{Depreciated Book Value: } B(t) = C_0 \times \max(0.15,\ 1 - t / L)$$

$$\text{Unfunded Capital Deficit: } \Delta(t) = R(t) - B(t)$$

C. Dynamic Equity Amortization & Workforce Cost-Per-Seat

$$\text{Ownership \%} = ((\text{Total Cost} - \text{Remaining Balance}) / \text{Total Cost}) \times 100$$

$$\text{Cost / Seat} = \text{Total Department Allocated Hardware Value} / \text{Active Headcount}$$

D. Green IT Power & ESG Carbon Conversion

$$\text{Monthly Energy (kWh)} = (\sum P_{watts} \times 9 \text{ hrs/day} \times 22 \text{ days}) / 1000$$

$$\text{Carbon Emissions (kg } CO_2\text{)} = \text{Monthly kWh} \times 0.71 \text{ kg } CO_2/\text{kWh}$$

6. EXECUTION ENVIRONMENT & DEPLOYMENT INSTRUCTIONS

Execute the following commands from the project directory to deploy the integrated pipeline:

```
      # Step 1: Initialize Database Engine (Port 3306) & "C:\Program Files\MySQL\MySQL
Server 8.0\bin\mysql.exe" -u root -p -e "USE corporate_asset_management; SHOW TABLES;" # Step
2: Boot Spring Boot Application Gateway (Port 8080) cd "D:\New folder (4)\backend" &
".\.maven\apache-maven-3.9.9\bin\mvn.cmd" spring-boot:run # Step 3: Launch React 18 SPA User
Interface (Port 5173) cd "D:\New folder (4)\frontend" npm.cmd run dev --port 5173
```